GUARDIAN

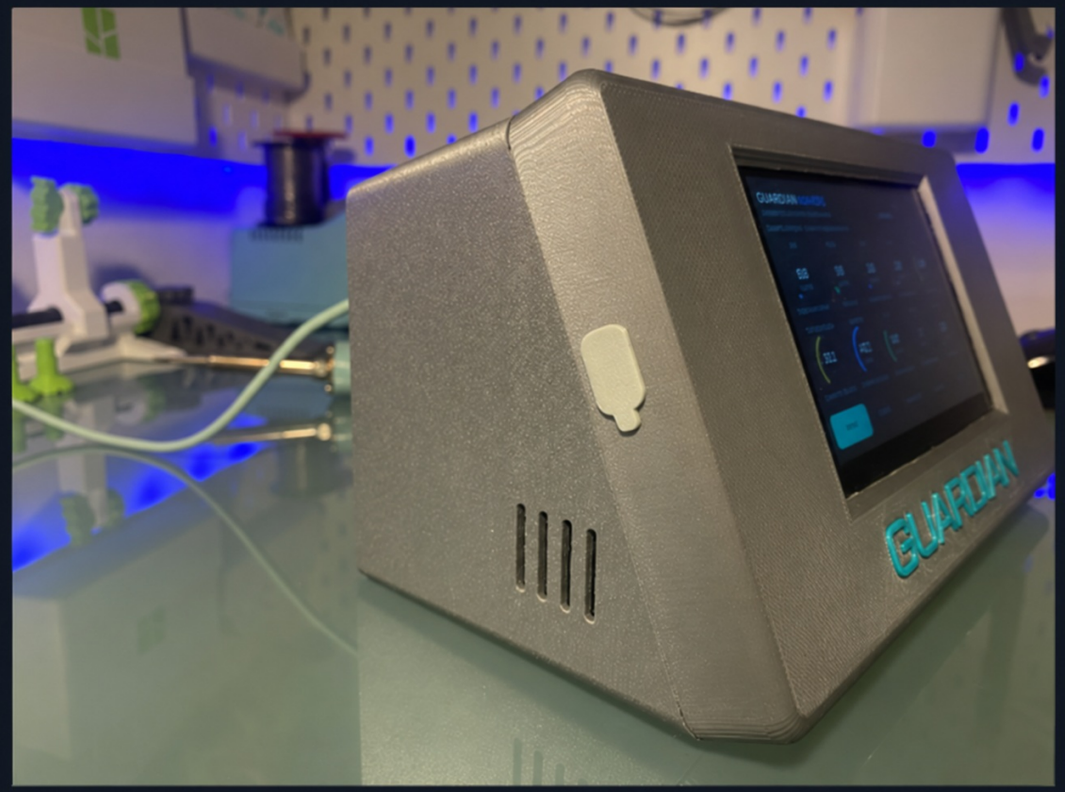
MAKERS

The environment during 3D printing

Environmental monitoring
for maker workspaces

Product presentation
October 2026

Designed for the workbench



Observe

Follow environmental changes while working and compare different moments within the same session.

Document

Save readings in CSV files alongside print status and filament information reported by the printer.

Interpret with care

GUARDIAN is an environmental monitor, not a reference instrument, a safety detector or proof of healthy air.

Ten parameters in one overview

Parameter	Unit	Interpretation
PM1 / PM2.5 / PM4 / PM10	µg/m³	Optically estimated mass concentrations by particle-size fraction.
CO2	ppm	Helps interpret ventilation and room occupancy.
Temperature / humidity	°C / % RH	Personal comfort ranges, separate from alerts.
VOC / NOx	Index	Relative indicators, not concentrations of individual substances.
HCHO	ppb	Formaldehyde; a 30-minute average is also available.

Integrated Sensirion SEN69C sensor. Specifications and limitations: S1, S4, S19. It does not directly measure the ultrafine particles typical of some 3D-printing emissions.

HOME: readings at a glance

Values and units

Ten panels bring together measurements, indices and status indicators.

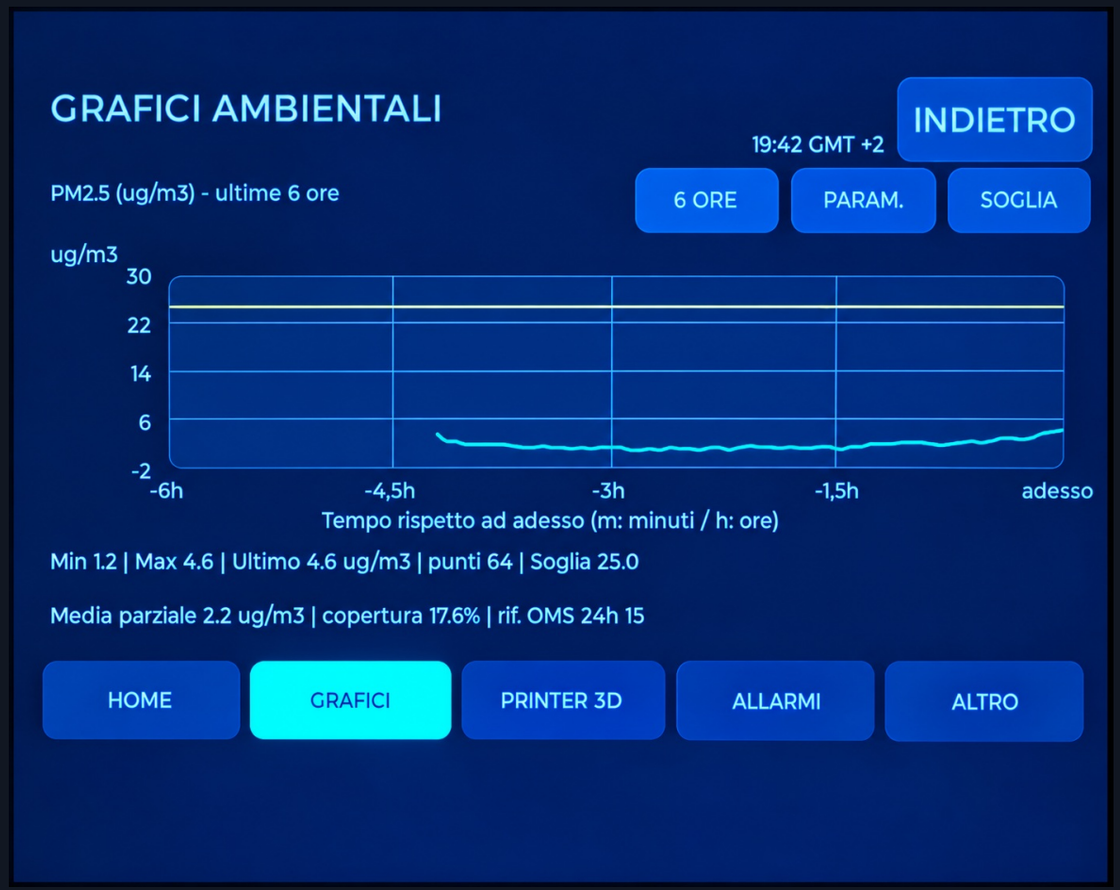
Thresholds and comfort

Personal thresholds are separate from temperature and humidity comfort ranges.

Simple navigation

HOME, CHARTS, PRINTER 3D, ALERTS and OTHER remain accessible.

Charts: following the trends



Time windows

2, 6, 12 and 24-hour views, with a light grid and integer labels on the vertical axis.

Information retained

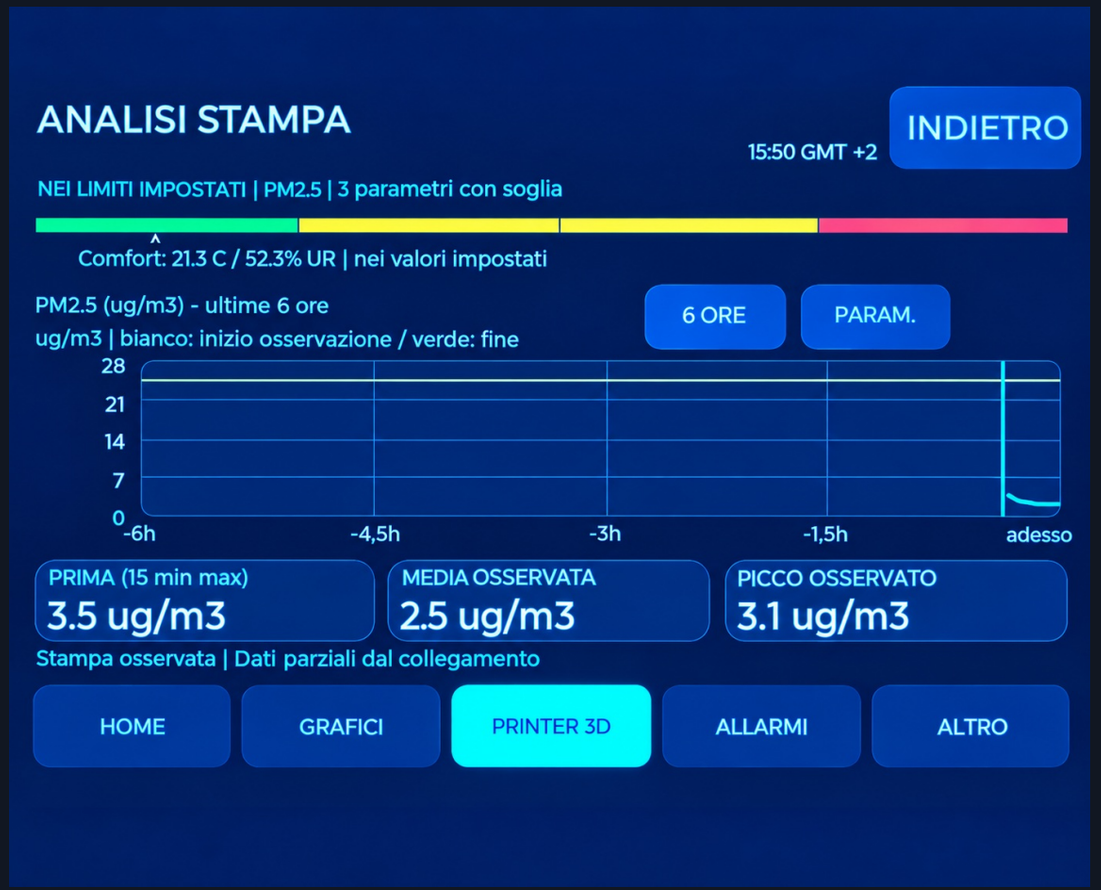
The current reading, statistics and reference information remain below the chart.

Relevant averages

PM2.5 and PM10: 24 hours. HCHO: 30 minutes. Incomplete periods are identified.

Chart history uses volatile memory; microSD provides persistent recording. Averaging does not make the device a regulatory instrument.

Printing in its environmental context



Before, during and peak

Compare the selected parameter with the preceding period and the session readings.

Overall status

A bar summarises assessable parameters against the configured thresholds.

Interpreting a correlation

Simultaneous changes do not prove the printer is the only cause.

A session started after printing began, or containing data gaps, is identified as partial.

The complete PRINTER 3D page

Tested printer

One Bambu Lab on the local network.
A1 tested; X1 Carbon connectivity verified.

Print information

Preview, progress, layers, remaining time, temperatures and filament, when available.

Data continuity

After a disconnection, retained readings remain distinguishable from fresh data.

The connection is for monitoring: GUARDIAN does not stop the printer. Preview and job name depend on the data and file available on the printer.

Alerts and comfort serve different purposes

Alert centre

Collects events and shows threshold exceedances and conditions requiring attention.

Personal thresholds

User-defined operating values are not certified health limits.

Comfort indication

Temperature or humidity outside the personal range generates a comfort indication.

A normally switched-off or idle printer is not treated as a fault. Loss of data during a print is a separate condition.

Viewing beyond the device

PC and smartphone

Browser access on the local network; remote access through your network VPN.

Configuration

Network, time, lighting and the settings available in the web interface.

Device identification

The NETWORK page shows the device address and numerical Wi-Fi signal strength.

Anonymised photograph with example details. The local web service uses HTTP: do not expose it directly to the Internet.

Lighting and time settings

Manual

Adjust brightness from the display or smartphone browser.

Automatic

Day/night brightness with start and stop times, including periods crossing midnight.

Time zone

Select a time-zone profile; daylight saving follows the selected profile.

Keeping a record of the data

microSD

Periodic CSV recording with units and session context.

Browser download

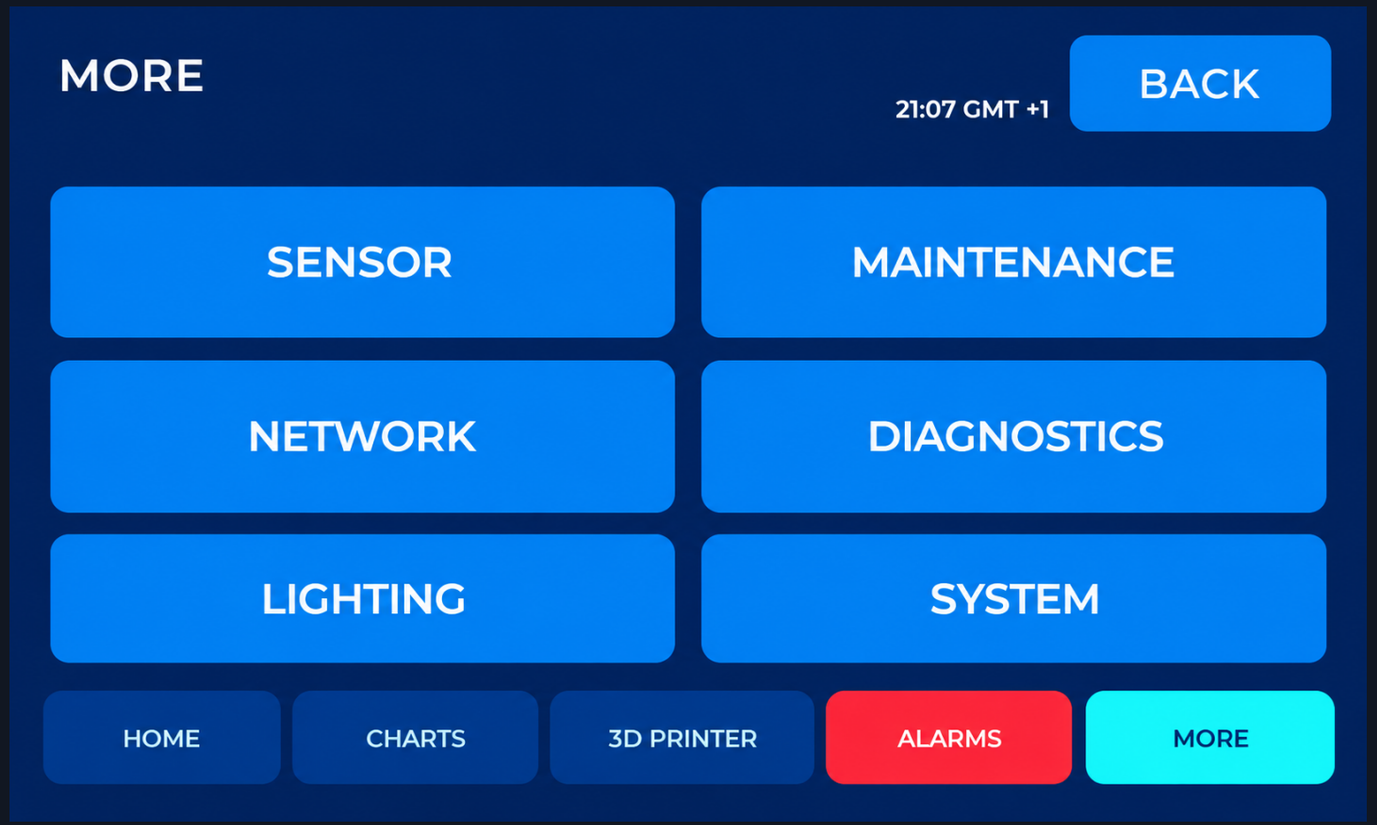
A dedicated current-CSV link and subsequent pages for historical files.

Local management

Read, test and format the card, with explicit confirmation.

FORMAT erases every file on the microSD card.

A clear technical workflow



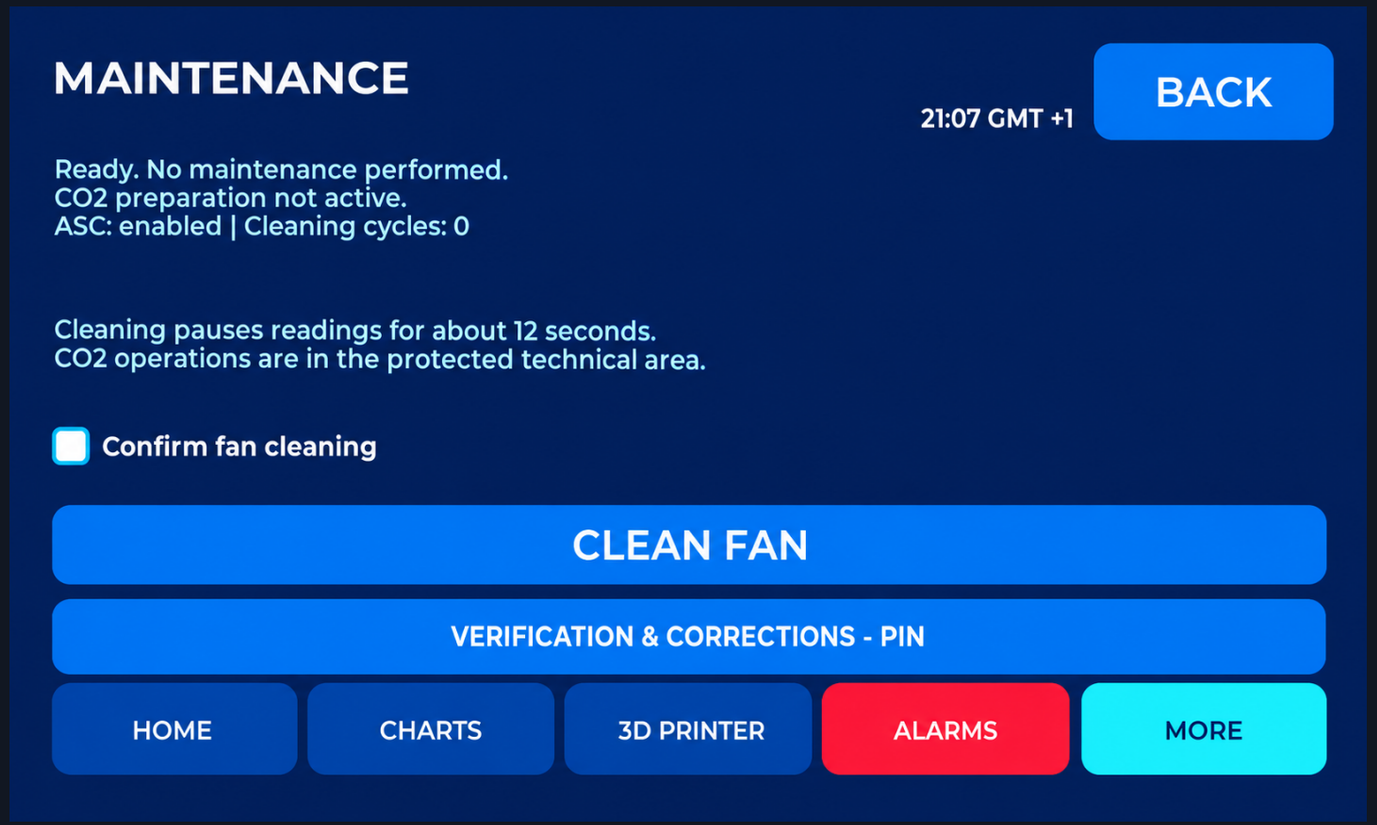
Cleaned demonstration screen based on a real photograph.

Open MORE → MAINTENANCE → VERIFY & CORRECT.

First check SEN69C presence and status on the SENSOR page.

This edition describes GUARDIAN 1.1.0 and the functions tested in build TEST4 FIX1.

Maintenance and preliminary checks



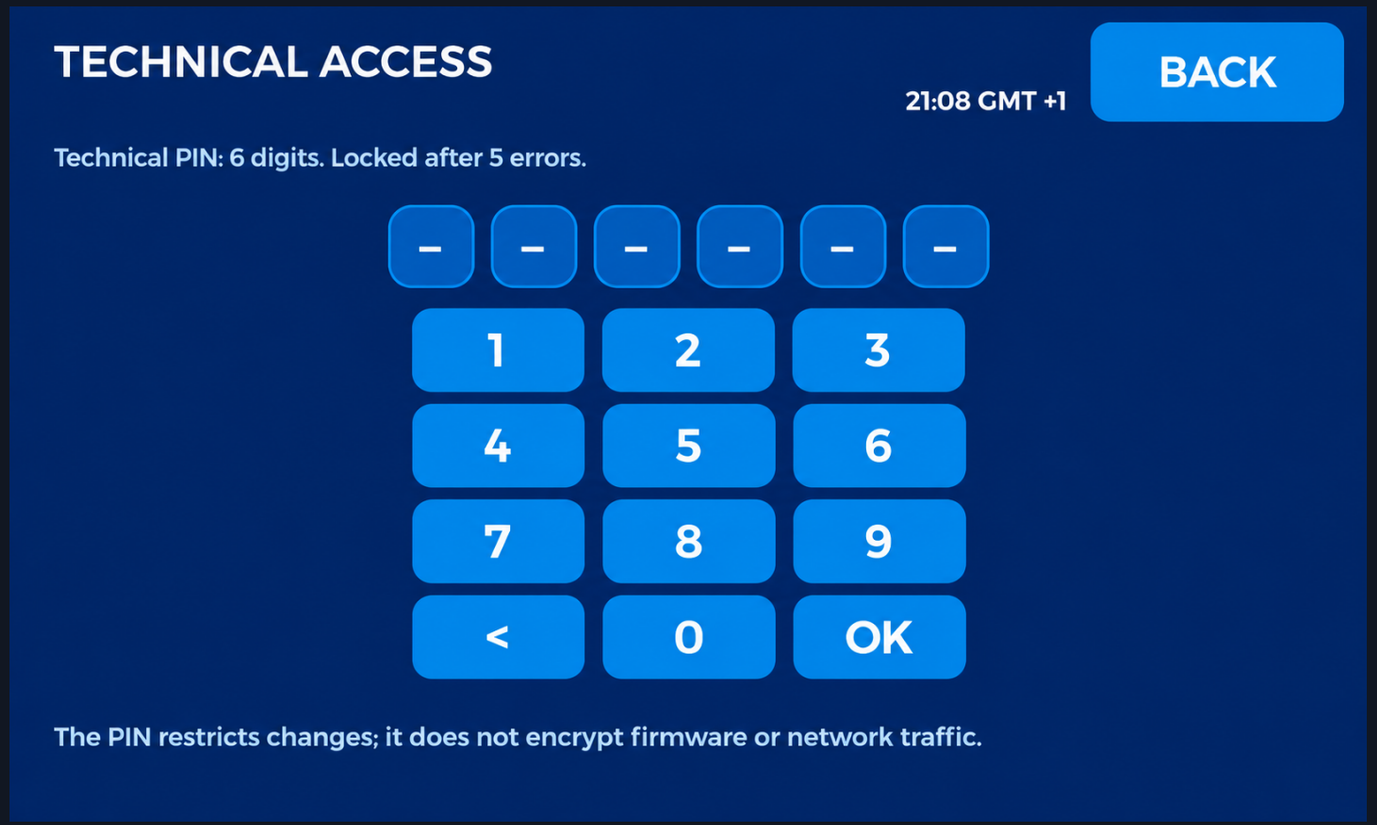
Cleaned demonstration screen based on a real photograph.

Fan cleaning is a sensor maintenance operation.

Before comparing instruments, allow stabilisation and record placement, environmental conditions, sampling interval and reference instrument.

Compare the same quantity and unit.

Technical access: six-digit PIN



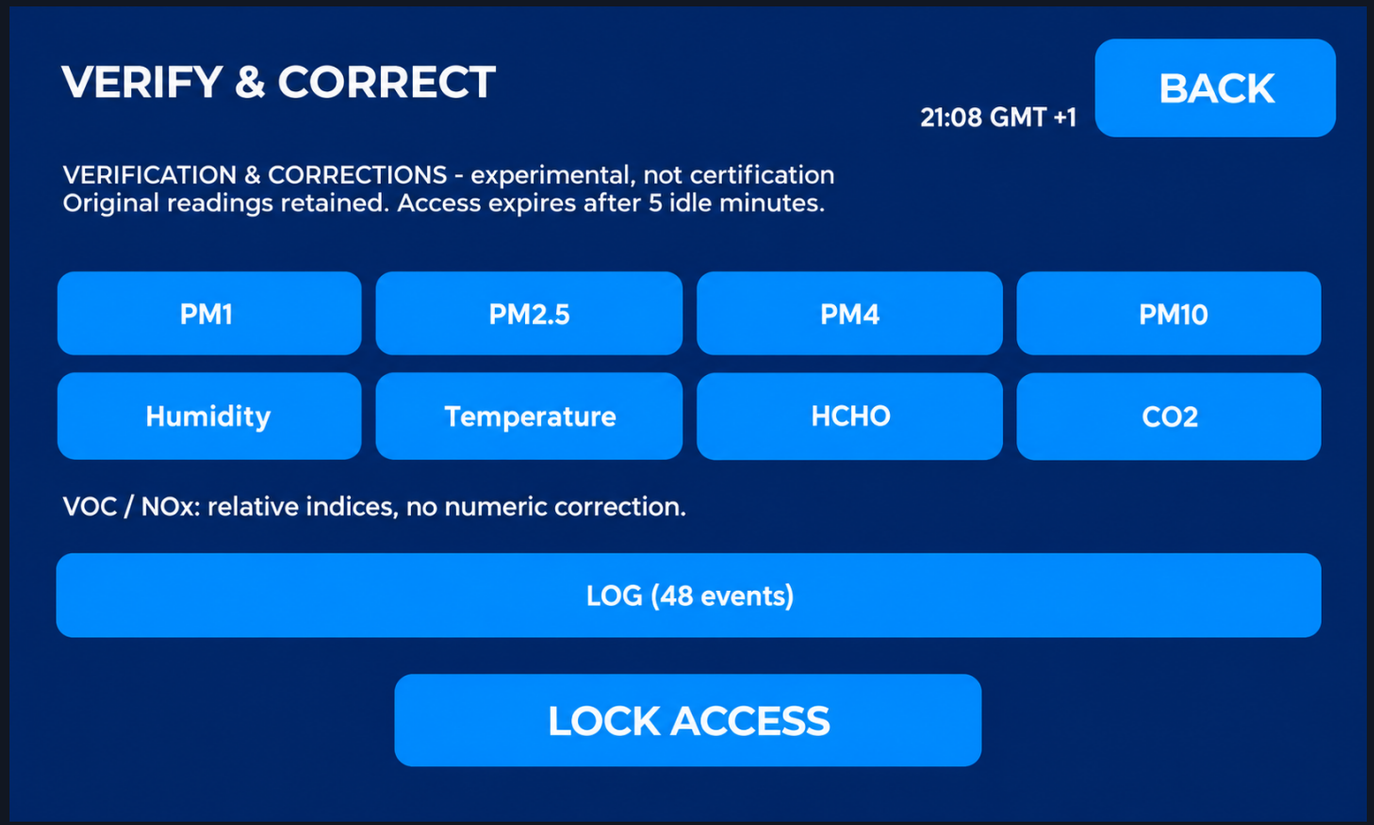
On first access, choose a six-digit PIN and enter it twice.

After five incorrect attempts, wait 60 seconds before retrying.

The PIN restricts interface changes; it does not encrypt memory or enable Secure Boot.

Cleaned demonstration screen based on a real photograph.

Which parameters can be corrected



PM1, PM2.5, PM4, PM10, temperature, relative humidity, HCHO and CO2 are available.

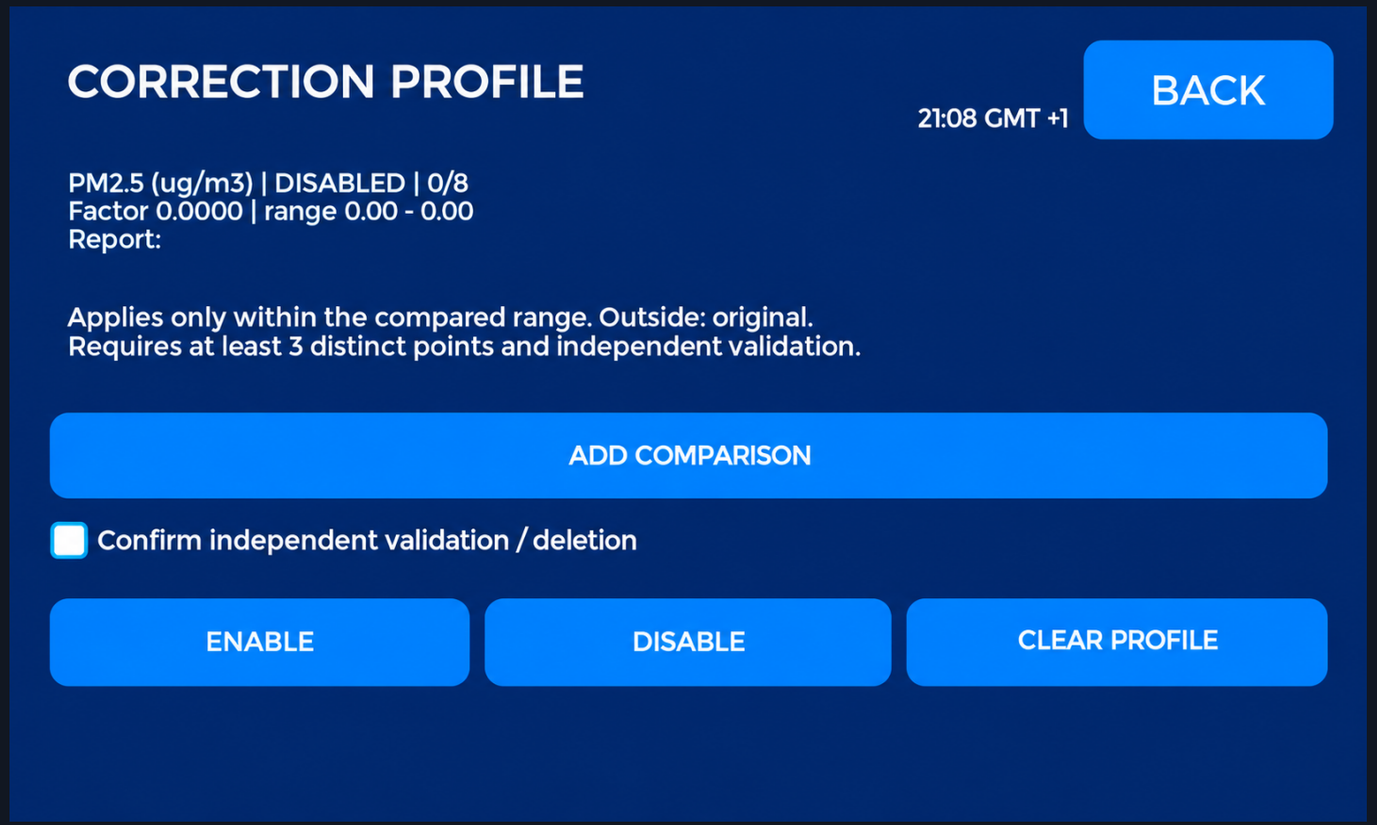
Each parameter has a profile containing comparisons, factor or offset, valid range and report reference.

Particulate channels use a multiplicative factor; the other channels use an additive offset.

Cleaned demonstration screen based on a real photograph.

DEMO/2026 · Example, not certification

Profile: enabling and valid range



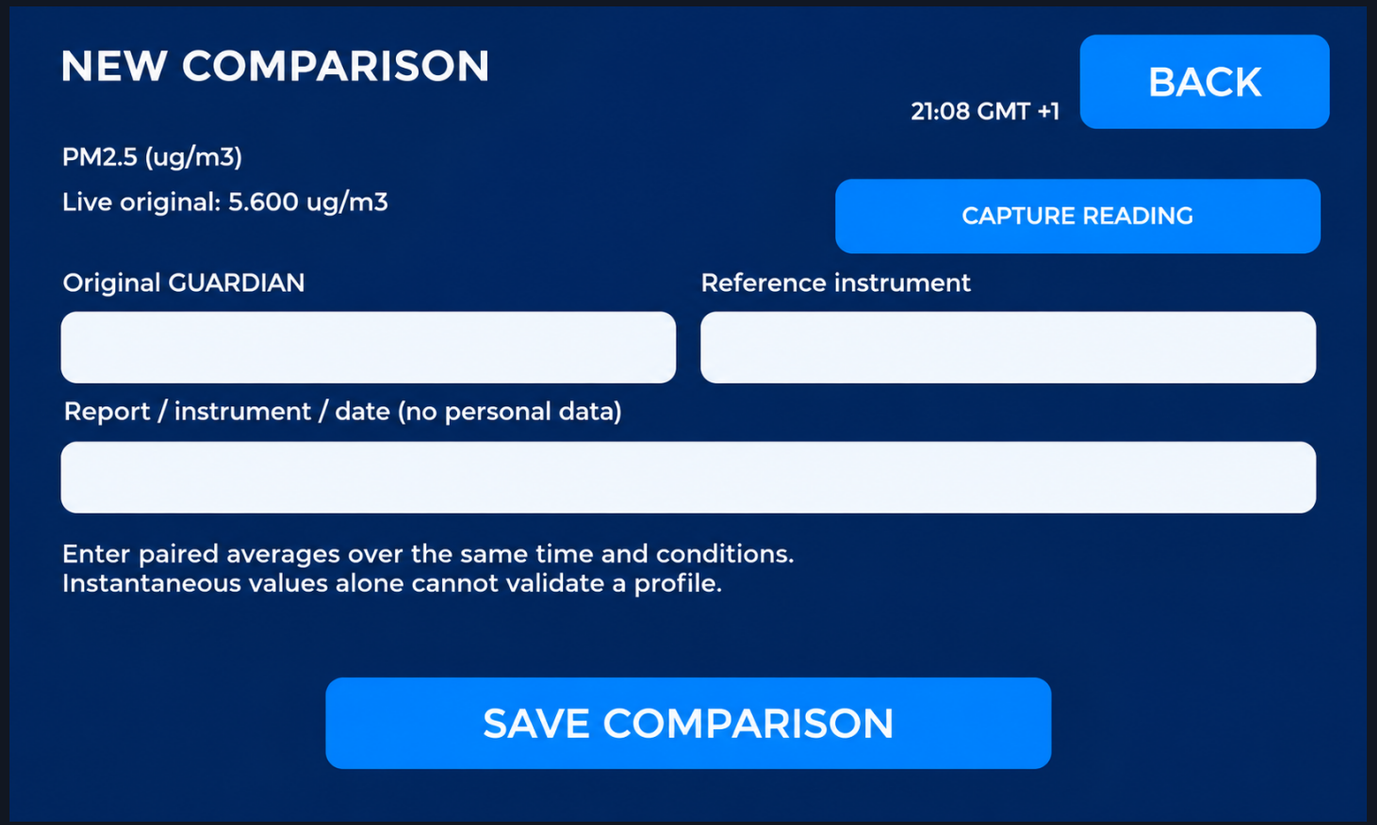
Cleaned demonstration screen based on a real photograph.

Enabling a correction requires at least three distinct original readings, a report identifier and confirmation of independent verification.

Correction applies only within the range of the compared original readings.

DISABLE retains comparisons but suspends correction.

New comparison: original and reference



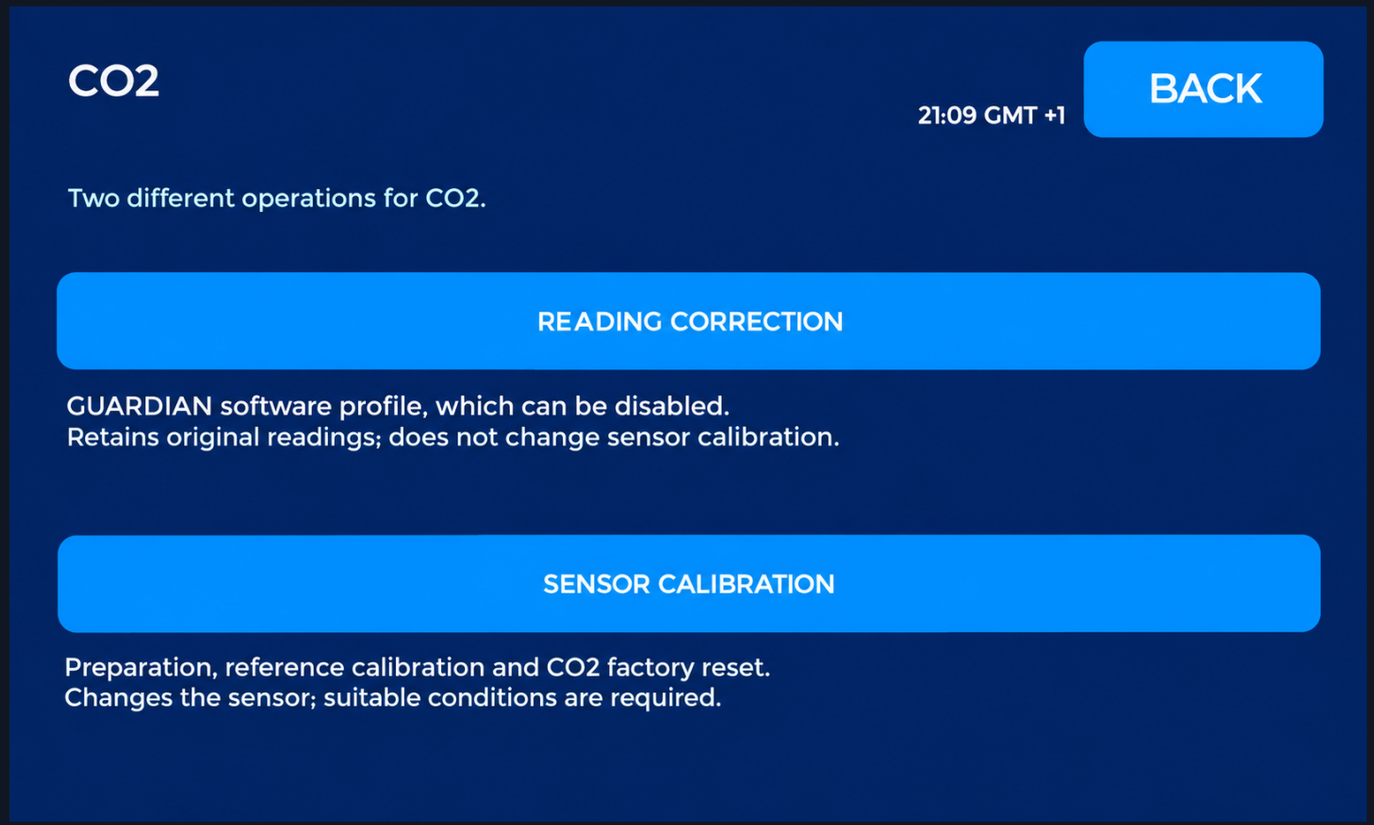
GUARDIAN ORIGINAL shows the current reading before GUARDIAN software correction.

Capture the reading into the comparison field, then enter the reference instrument reading in the same unit.

The report field accepts /, for example DEMO/2026.

Cleaned demonstration screen based on a real photograph.

CO2: two different operations



Cleaned demonstration screen based on a real photograph.

The CO2 entry groups the software correction profile and internal sensor operations.

Software correction adds an offset within the verified range and can be disabled.

Do not stack the two operations: disable the software CO2 correction before internal calibration.

CO2 example: one point is not enough

The screen shows one comparison: an original reading of 887 ppm and a calculated +13 ppm offset.

The software enabling requirements are not yet met.

Select points representative of intended use within a suitable comparison plan.

Cleaned demonstration screen based on a real photograph.

CO2 comparison and data retention

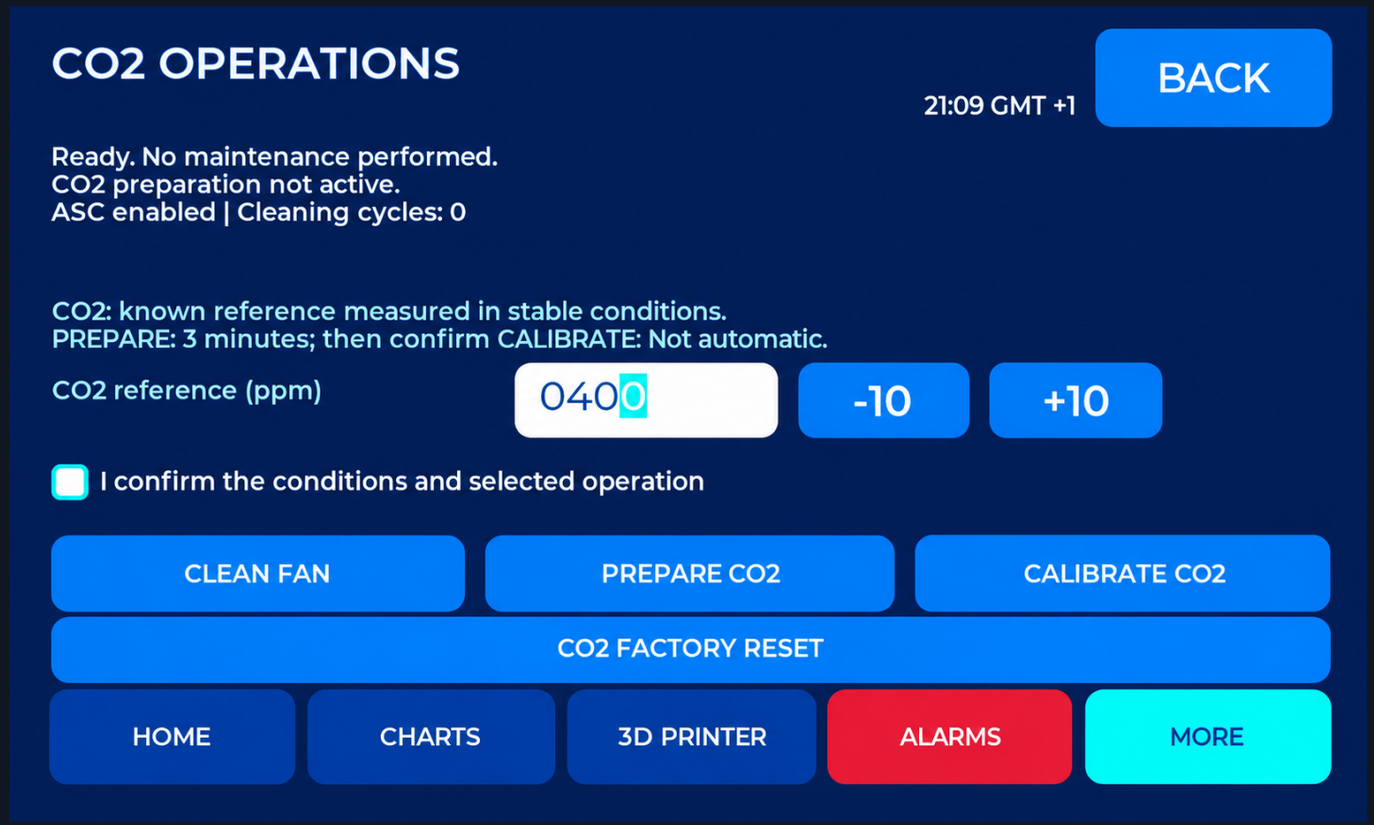
Cleaned demonstration screen based on a real photograph.

The original reading continues updating; the captured comparison value remains the selected value.

Profiles and the technical log are stored on microSD.

Restarting preserves the PIN and profiles.

CO2 operations: prepare and verify



Cleaned demonstration screen based on a real photograph.

Use a reliable, known reference within the allowed 400–2000 ppm range.

PREPARE starts a three-minute wait.

CO2 RESET concerns the sensor channel, not the whole GUARDIAN.

Technical log and traceability

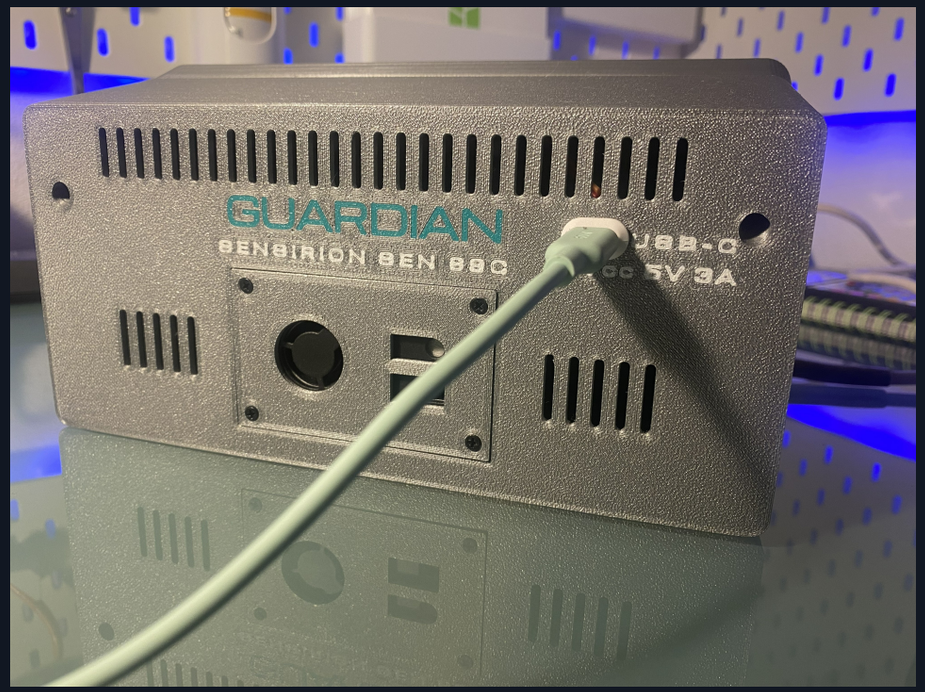
Log: new optimized photo (Italian display).

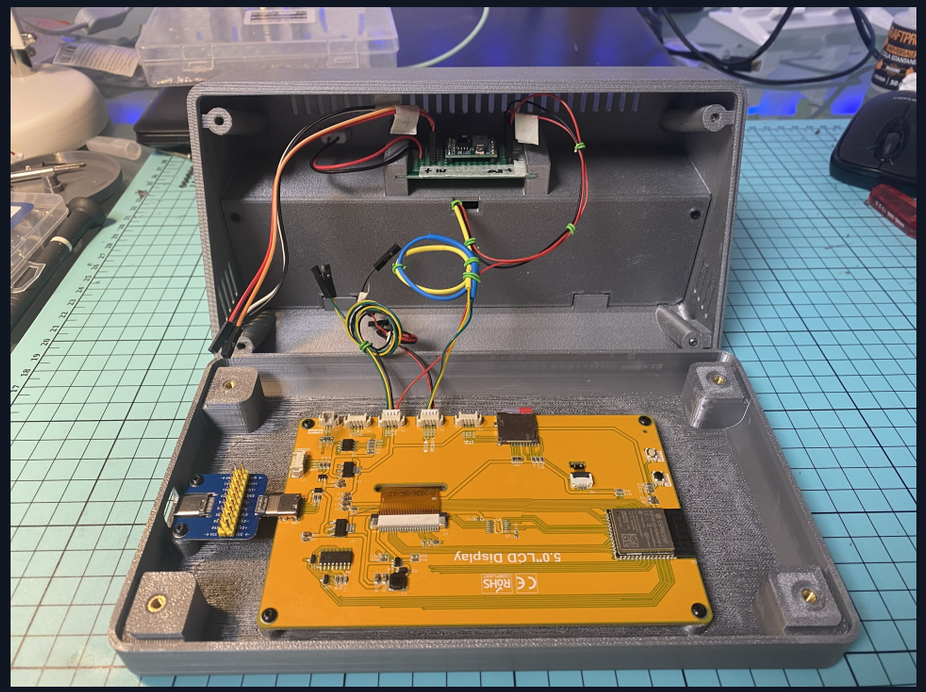
The log retains the last 48 events: comparison additions, profile enabling, disabling and clearing.

To export: open the browser on the same network, MAINTENANCE > Verification & corrections (PIN).

Columns show sequence, UTC, uptime, parameter, action, origin, before/after and report.

Accessible and serviceable





Two USB-C access points

Rear socket for power; side socket for programming, protected by a TPU cap.

Sensor and electronics compartments

Rear-mounted SEN69C, removable support and internal partition. Keep the airflow paths clear and separate.

Understanding the references

Indication	Reference	Correct use
PM2.5	WHO 2021: 15 µg/m³ / 24 h	Complete after 24 hours of valid recording; earlier averages are partial.
PM10	WHO 2021: 45 µg/m³ / 24 h	Complete after 24 hours of valid recording; earlier averages are partial.
HCHO	WHO 2010: 0.1 mg/m³ / 30 min	About 81 ppb at 25 °C and 1 atm; conversion depends on conditions.
CO2, VOC, NOx	Ventilation / relative indices	They do not detect every pollutant or establish safe air on their own.

S6, S7, S10. WHO guidelines do not certify GUARDIAN or replace assessment of occupational exposure.

Interpreting measurements correctly

Continuous monitoring

GUARDIAN combines environmental readings, history, microSD recording, web access and printer information in one interface.

Accuracy and operating conditions

Readings also depend on placement, ventilation and thermal conditions. Independent metrological validation of the complete device is not available.

Limits that remain relevant

Optical PM does not directly measure the entire ultrafine fraction. Gas indices do not identify every compound. Low readings do not automatically mean safe air. [S1, S8, S9]

Calibration and conformity

Sensirion statement

Sensirion states that each device in the listed families, including SEN69C, is individually tested after calibration. The statement covers SEN5x/SEN6x particulate-matter sensors; the original is attached to the manual. It is not an individual GUARDIAN calibration report or certification of all its channels. [S2]

The complete product

Component REACH/RoHS documents contribute to the technical file; they are not a declaration of conformity for the whole device. [S16]

Before any commercial distribution

Determine the obligations applicable to the finished product and assess safety, radio/EMC and documentation. RED, RoHS, WEEE and REACH are references to assess; no completed certification is claimed here. [S11–S15]

Display lighting: validated update

25 kHz backlight PWM

The owner confirmed that the 5 kHz whine disappeared after the change to 25 kHz. Manual brightness and the automatic day/night schedule were tested successfully.

Version identification

This edition describes bilingual GUARDIAN with the 25 kHz modification. The tested sketch is labelled 1.0.0-rc3-pwm25-test1; this identifier may still appear on the device and in the browser.

Settings and updates

Brightness remains adjustable from the display and browser. This change requires neither a factory reset nor microSD formatting.

Scope of testing

Functional checks on the owner’s prototype support retaining this setting. They do not constitute metrological validation or certification of the complete product.

The OTHER menu

Select SYSTEM (SISTEMA) to choose the language or access restart and reset.

Retouched from the supplied photograph: reflections removed and perspective corrected. Original labels and controls retained.

System and interface language

Choose Italiano or English: the language is saved without restarting. RESTART keeps settings; FACTORY RESET clears GUARDIAN settings and web access. CSV files remain on microSD.

Retouched from the supplied photograph: reflections removed and perspective corrected. Original labels and controls retained.

Italian and English on the same device

Selection on the display

OTHER → SYSTEM: choose Italiano or English. The choice is saved without restarting. With no saved preference, the initial language is Italian.

PC and smartphone

They follow the display language by reloading within about three seconds while reachable. Save open forms before changing language.

After restarting

The choice is retained. A GUARDIAN factory reset restores Italian. Job names, user-entered data and the CSV format remain unchanged.

Real readings and compatibility

24 hours for a complete average

A complete PM2.5 / PM10 average requires 24 hours of valid recording. Earlier averages cover only the available period and are partial. HCHO uses a separate 30-minute period.

Photographs and demonstrations

Display photographs show real readings collected over partial periods, not complete 24-hour averages. Network details and credentials are examples. Firmware reconstructions are identified separately.

Printer used for testing

A1 tested; X1 Carbon connectivity verified. GUARDIAN uses Bambu Lab protocols; other models require verification. Other brands require investigation and possible adaptation.

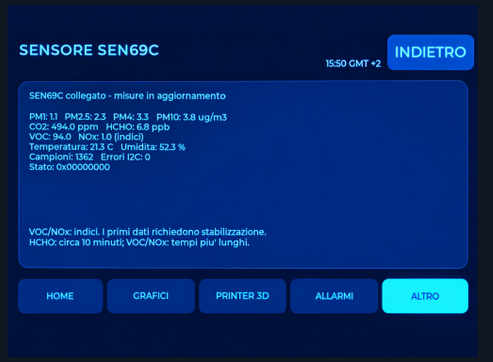
Bilingual edition

This documentation describes the Italian and English interface. The owner reports successful initial functional testing; software testing is not metrological validation.

Other service screens

OTHER

Access to additional functions.



SENSOR

Module presence and status information.

DIAGNOSTICS

Display, touch and SD tests.

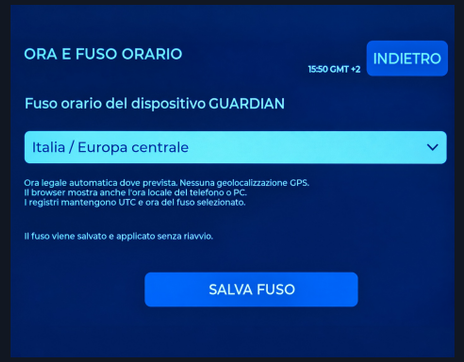
SD CARD

Recording status and local commands.

Controls and configuration

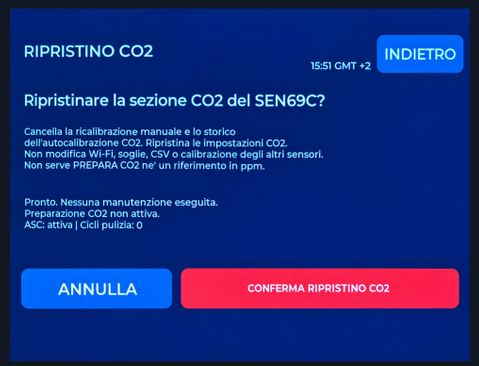
PARAMETER THRESHOLD

Set a personal operating threshold.



TIME AND TIME ZONE

Geographical clock profile.



CO2 RESET

Separate confirmation for the CO2 channel.

SYSTEM

Restart or reset GUARDIAN.

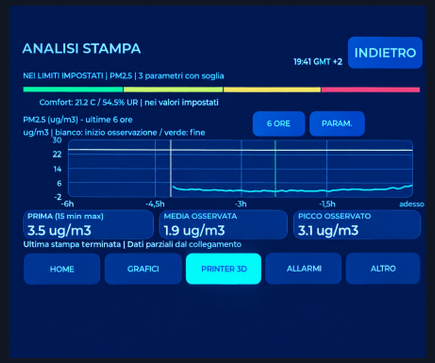
Recognising environmental states

HOME: CO2 alert

Colour indicates the state against the configured threshold.

ALERT CENTRE

Read the cause and current value; alerts are informational.



ANALYSIS: print completed

Session summary and available samples.

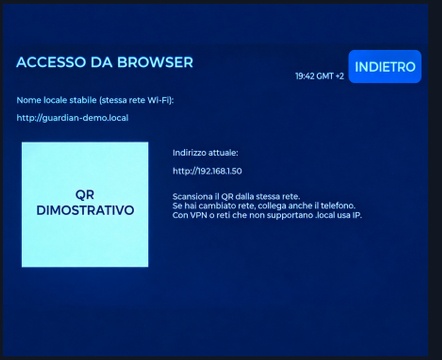
FORMAT: confirmation

Erases all card files; back up first.

Access and local checks

PRINTER SETTINGS

Your printer?s IP, serial number and local code. Example details shown.



BROWSER ACCESS

Use your monitor’s address. The QR code shown was replaced.

DISPLAY TEST

Check uniformity and persistent defects.

TOUCH TEST

Check response at different screen positions.

Adjusting thresholds quickly

Small or large steps

±1, ±10 and ±100 buttons: increase by 100 with one tap on +100.

Press and hold

Repeats the change; holding ±1 accelerates the step.

Apply and retain

Check the command result. In the browser, enter the value directly.

Personal operating thresholds are not certified health limits.

Display and browser

An organised sequence

ENVIRONMENTAL TRENDS → ENVIRONMENTAL AVERAGES; PRINT ANALYSIS → 3D PRINTING on PC and smartphone pages.

Readable job names

“tappo in tpu + tappo in tpu” appears as “tappo in tpu”. Different names remain visible. The original print file is not renamed.

MAINTENANCE

The menu uses the full label. CO2 preparation, calibration and cleaning remain separate procedures.

Data and images

The model preview is available on the monitor. Screens in this guide are examples; network details, credentials and serial numbers were replaced with demonstration values.

Sources and reference documents

S1 · Sensirion — SEN6x datasheet (v1.0, October 2026)
Open original document

S2 · Sensirion — SEN5x / SEN6x factory calibration
Open original document

S3 · Sensirion — SEN6x mechanical integration
Open original document

S4 · Sensirion — specification statement
Open original document

S5 · Sensirion — temperature compensation
Open original document

S6 · WHO — Global Air Quality Guidelines, 2021
Open original document

S7 · WHO — Indoor Air Quality: selected pollutants, 2010
Open original document

Accessed: 13 October 2026. Component documents do not certify the complete GUARDIAN device. The photographs show example readings, not metrological validation results.

Sources and reference documents

S8 · EPA — research on 3D-printing emissions
Open original document

S9 · EPA — Air Sensor Toolbox
Open original document

S10 · UBA — CO2 and ventilation
Open original document

S11 · Radio Equipment Directive 2014/53/EU
Open original document

S12 · RoHS Directive 2011/65/EU
Open original document

S13 · Delegated Directive (EU) 2015/863 — RoHS
Open original document

S14 · WEEE Directive 2012/19/EU
Open original document

Accessed: 13 October 2026. Component documents do not certify the complete GUARDIAN device. The photographs show example readings, not metrological validation results.

Sources and reference documents

S15 · REACH Regulation (EC) 1907/2006
Open original document

S16 · Sensirion — REACH / RoHS statement
Open original document

S17 · Espressif — Arduino ESP32 installation
Open original document

S18 · Freenove — FNK0115 repository
Open original document

S19 · Sensirion — SEN69C product information
Open original document

Accessed: 13 October 2026. Component documents do not certify the complete GUARDIAN device. The photographs show example readings, not metrological validation results.